TITLE: AI-ASSISTED SMART DENTAL SUCTION SYSTEM CAPABLE OF REAL-TIME MONITORING OF SALIVA POOLING, SUCTION PERFORMANCE, AND PATIENT HEAD POSITION TO ENHANCE AIRWAY SAFETY DURING DENTAL PROCEDURES

ABSTRACT

Evaluation of an AI-Assisted Smart Dental Suction System for Real-Time Airway Safety Monitoring during Dental Procedures is an intelligent healthcare application designed to improve patient safety and clinical efficiency during dental treatments. The system continuously monitors saliva pooling, suction performance, and patient head position using Artificial Intelligence (AI), embedded sensors, and real-time monitoring technologies. The proposed system helps dentists and healthcare professionals maintain effective airway management and reduce the risk of saliva accumulation, choking, and aspiration during dental procedures.

The application utilizes smart suction monitoring techniques integrated with sensors such as flow sensors, saliva level detectors, and head position monitoring modules. The AI-based system analyzes suction efficiency, detects abnormal saliva accumulation, and evaluates patient head orientation in real time. Based on the collected data, the system provides instant alerts and automated monitoring support to assist dental practitioners in maintaining proper suction performance and patient airway safety throughout the procedure.

The proposed system also enables continuous monitoring through a digital interface that displays suction status, saliva levels, and patient positioning information. The intelligent monitoring mechanism improves procedural safety by reducing manual observation requirements and supporting faster clinical response during emergencies. Additionally, the system stores monitoring records digitally for future analysis and treatment documentation.

Compared to conventional dental suction systems, the proposed AI-assisted smart dental suction system combines airway safety monitoring, intelligent suction analysis, patient position tracking, and real-time healthcare assistance into a single integrated platform. The primary objective of the system is to enhance airway protection, improve suction efficiency, support safer dental procedures, and provide intelligent monitoring assistance for both patients and dental professionals.

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FIELD OF THE INVENTION

This invention pertains to the field of dental healthcare technology, artificial intelligence, and smart patient monitoring systems, specifically relating to AI-assisted dental suction management and real-time airway safety monitoring during dental procedures using intelligent sensor-based systems.

BACKGROUND OF THE INVENTION

Traditional dental suction systems mainly depend on manual operation and continuous observation by dental professionals, which may sometimes lead to inefficient saliva removal, improper suction performance, and increased risk of saliva pooling during treatment procedures. Patients undergoing dental procedures may experience discomfort, choking risk, airway obstruction, or aspiration problems due to inadequate suction monitoring and improper patient head positioning.

Existing dental suction systems primarily focus only on saliva removal and lack intelligent monitoring capabilities for evaluating suction efficiency, saliva accumulation, and patient positioning in real time. Continuous monitoring of airway safety during dental procedures requires constant attention from healthcare professionals, making the process difficult and time-consuming in busy clinical environments.

Additionally, current dental monitoring applications lack integrated AI-based airway safety analysis, automated suction performance evaluation, and intelligent alert mechanisms. Dental professionals require advanced digital systems capable of monitoring patient safety conditions continuously and providing immediate alerts during abnormal situations.
Therefore, there is a need for an advanced healthcare platform that integrates Artificial Intelligence (AI), real-time saliva pooling detection, suction performance analysis, patient head position monitoring, and intelligent healthcare assistance into a single smart system to improve airway safety and enhance dental treatment efficiency.

SUMMARY OF THE INVENTION

The invention is an AI-assisted smart dental suction system designed for enhancing airway safety during dental procedures. The system continuously monitors saliva pooling, suction efficiency, and patient head position using Artificial Intelligence, embedded sensors, and real-time monitoring technologies.

The system provides real-time monitoring reports, suction performance feedback, and intelligent safety assistance for both patients and healthcare professionals. It also supports future enhancements such as cloud-based patient monitoring, automated dental healthcare reporting, and advanced AI-driven clinical safety management systems.

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SPECIFICATION

- The invention relates to an Artificial Intelligence (AI)-assisted smart dental suction system designed to enhance airway safety during dental procedures by monitoring saliva pooling, suction performance, and patient head position using advanced sensor technologies, embedded systems, and intelligent monitoring algorithms.

- Patients undergoing dental treatments often experience saliva accumulation, discomfort, choking risk, and airway obstruction during lengthy procedures. Conventional dental suction systems mainly rely on manual operation and continuous observation by dental professionals, which may lead to inconsistent suction performance and delayed response during emergency situations. Existing dental systems provide limited support for intelligent airway safety monitoring, automated suction analysis, and real-time patient position evaluation.

- The invention is an AI-assisted dental healthcare system integrated with real-time monitoring sensors, intelligent processing modules, and digital healthcare support mechanisms. The system continuously monitors saliva levels, evaluates suction efficiency, and tracks patient head orientation during dental procedures. The AI-based monitoring mechanism analyzes sensor data and provides instant safety alerts, abnormal condition notifications, and continuous airway monitoring assistance for dental professionals..

- The proposed system includes intelligent suction monitoring, saliva pooling detection, patient safety analysis, and digital healthcare reporting features. The system stores monitoring records securely within a digital database and provides healthcare professionals with an intelligent dashboard to monitor treatment safety and patient conditions effectively during dental procedures.

- The invention further supports advanced dental healthcare management through real-time monitoring, embedded sensor integration, intelligent alert systems, and AI-powered safety analysis. Future enhancements may include cloud-based dental monitoring services, automated healthcare reporting, remote clinical supervision, voice-enabled healthcare assistance, and advanced predictive analytics for improving patient airway safety and overall dental treatment efficiency.

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DESCRIPTION

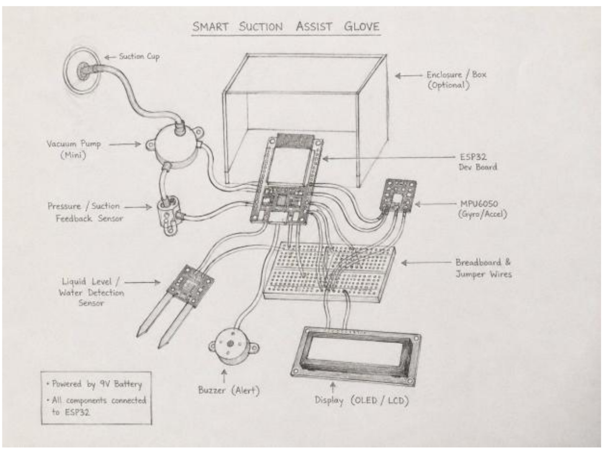
AI-Assisted Smart Dental Suction System for Real-Time Airway Safety Monitoring is an intelligent healthcare application that uses Artificial Intelligence (AI), sensor-based monitoring, and embedded healthcare technologies to evaluate saliva pooling, suction performance, and patient head position during dental procedures.

The system continuously monitors saliva accumulation and suction efficiency using smart sensors and intelligent monitoring mechanisms. It also tracks patient head orientation to ensure proper airway safety during treatment procedures. Based on the real-time analysis, the application generates safety alerts, suction performance feedback, and airway monitoring assistance for dental professionals.

The application also stores monitoring records digitally, enabling dentists and healthcare professionals to monitor treatment conditions effectively through an intelligent healthcare platform.

Figure 1: System Architecture of Smart AI Dental Suction System

Figure 2: 2D Design for Smart AI Dental Suction System



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We claimed:

1. An AI-assisted smart dental suction system that enhances airway safety during dental procedures through real-time monitoring of saliva pooling, suction performance, and patient head position.
2. A sensor-based intelligent dental healthcare application integrated with Artificial Intelligence (AI), real-time monitoring technologies, and automated alert mechanisms to improve patient safety and suction management during dental treatments.
3. An intelligent airway monitoring system that continuously analyzes suction efficiency, saliva accumulation, and patient positioning using embedded sensors and AI-based monitoring algorithms to support safe and effective dental procedures.

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